

智能系统设计实践

六足机器人避障控制 第二次进展汇报

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2025 年 10 月 13 日



基于 JupyterLab 的基本运动代码实现和调试

- 在六足机器人上部署成功了 JupyterLab 环境，实现了直接板上运行代码和调试。
- 参考资料成功实现了在 Python 环境中控制六足机器人的运动。
- 实现了六足机器人的前进、后退、左转、右转等基本运动。
- 调用实现了六足机器人的表演动作。

代码实现-库导入和机器人定义

Muto 类负责整个机器人的协调，是机器人顶层类。

```
# 导入库 import library
import cv2
import time
import threading

import ipywidgets.widgets as widgets
from IPython.display import display
```

```
from MutoLib import Muto
g_bot = Muto()
```

代码实现-机器人控制

```
# 按键按下事件处理 Key press event processing
def on_button_clicked(b):
    with output:
        print("Button clicked:", b.description)
    if b.description == Name_widgets['Reset'][g_ENABLE_CHINESE]:
        g_bot.reset()
    elif b.description == Name_widgets['Stretch'][g_ENABLE_CHINESE]:
        g_bot.action(1)
    elif b.description == Name_widgets['Greeting'][g_ENABLE_CHINESE]:
        g_bot.action(2)
    elif b.description == Name_widgets['Retreat'][g_ENABLE_CHINESE]:
        g_bot.action(3)
    elif b.description == Name_widgets['Warm_up'][g_ENABLE_CHINESE]:
        g_bot.action(4)
    elif b.description == Name_widgets['Turn_around'][g_ENABLE_CHINESE]:
        g_bot.action(5)
    elif b.description == Name_widgets['Say_no'][g_ENABLE_CHINESE]:
        g_bot.action(6)
    elif b.description == Name_widgets['Crouching'][g_ENABLE_CHINESE]:
        g_bot.action(7)
    elif b.description == Name_widgets['Stride'][g_ENABLE_CHINESE]:
        g_bot.action(8)
```

```
def on_slider_step(value):
    global g_step
    g_step = value
    if button_forward.icon == 'check':
        g_bot.forward(g_step)
    elif button_backward.icon == 'check':
        g_bot.back(g_step)
    elif button_move_left.icon == 'check':
        g_bot.left(g_step)
    elif button_move_right.icon == 'check':
        g_bot.right(g_step)
    elif button_turn_left.icon == 'check':
        g_bot.turnleft(g_step)
    elif button_turn_right.icon == 'check':
        g_bot.turnright(g_step)
    print("step:", value)

widget_step = widgets.interactive(on_slider_step, value=slider_step)
```

代码实现-机器人控制

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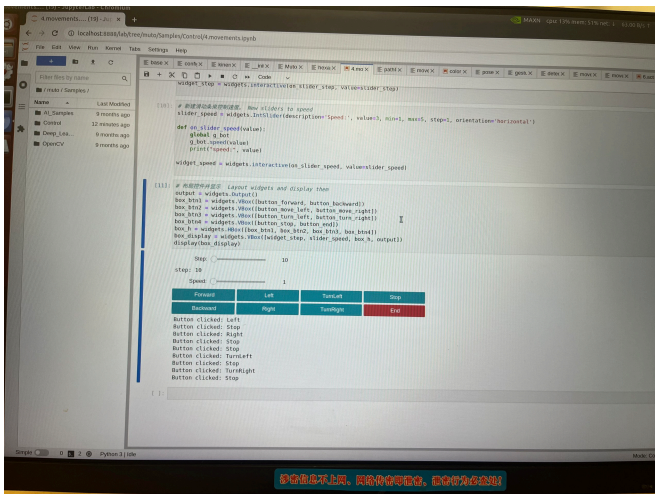
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智能系统设计实践

实现效果-基础控制

使用滑块空间来控制机器人的步长和速度。



涉密信息不上网，网络传密不保密，密件密件多处理！

视频展示

视频展示

- 对于底层代码进行修改，尝试改变机器人行走的步高等步态，实现自由控制。
- 调研相关文献资料，查询机器人足部控制相关代码。
- 实现机器人平稳上坡和下坡动作。
- 撰写中期报告。

Thanks!

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